MANIPAL
ACADEMY of HIGHER EDUCATION
(Institution of Eminence Deemed to be University)

MANIPAL SCHOOL OF INFORMATION SCIENCES

(A Constituent unit of MAHE, Manipal)

CYE 5103 Linux OS and Scripting – Class 1

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Part A — User and System Investigation (10 minutes)

Determine the username of the account you are currently using.

```
┌──(kali㉿kali)-[~]
└─$ whoami
kali
```

Display your UID, primary GID, and supplementary groups.

```
┌──(kali㉿kali)-[~]
└─$ id
uid=1000(kali) gid=1000(kali) groups=1000(kali),4(adm),20(dialout),24(cdrom),25(floppy),27(sudo),29(audio),30(dip),
44(video),46(plugdev),100(users),101(netdev),102(scanner),118(wireshark),119(kaboxer),968(vboxsf),982(bluetooth),99
9(lpadmin)

┌──(kali㉿kali)-[~]
└─$ groups
kali adm dialout cdrom floppy sudo audio dip video plugdev users netdev scanner wireshark kaboxer vboxsf bluetooth
lpadmin
```

Display all users currently logged in to the system.

```
┌──(kali㉿kali)-[~]
└─$ who
kali     seat0        2026-09-03 07:22 (:0)
```

Determine what activities the currently logged-in users are performing.

```
┌──(kali㉿kali)-[~]
└─$ w
 07:37:58 up 15 min,  1 user,  load average: 0.49, 0.54, 0.46
USER     TTY      FROM             LOGIN@   IDLE   JCPU   PCPU  WHAT
kali             -                07:22           0.00s   ?    lightdm --session-child 13 24
```

Examine /etc/passwd and identify the following information for your current account:

Username, UID, GID, Home directory, Login shell

```
┌──(kali㉿kali)-[~]
└─$ grep "^$(whoami):" /etc/passwd
kali:x:1000:1000::/home/kali:/usr/bin/zsh
```

Determine how many user-account entries are present in /etc/passwd.

```
┌──(kali㉿kali)-[~]
└─$ wc -l /etc/passwd
57 /etc/passwd
```

Display only the username and UID fields from /etc/passwd.

```
┌──(kali㉿kali)-[~]
└─$ cut -d: -f1,3 /etc/passwd
root:0
daemon:1
bin:2
sys:3
sync:4
games:5
man:6
lp:7
mail:8
news:9
uucp:10
proxy:13
www-data:33
backup:34
list:38
irc:39
_apt:42
nobody:65534
systemd-network:997
_dhcpcd:995
systemd-timesync:989
messagebus:988
tss:986
strongswan:100
tcpdump:985
sshd:984
_rpc:101
statd:102
dnsmasq:983
avahi:103
```

Find all accounts whose UID is 0. Explain the security significance of UID 0.

```
┌──(kali㉿kali)-[~]
└─$ awk -F: '$3 == 0 {print $1, $3}' /etc/passwd
root 0
```

Compare /etc/passwd and /etc/shadow. Why are password hashes stored in /etc/shadow instead of /etc/passwd?

```
┌──(kali㉿kali)-[~]
└─$ ls -l /etc/passwd /etc/shadow
-rw-r--r-- 1 root root   3256 Jun 16 10:30 /etc/passwd
-rw-r----- 1 root shadow 1464 Jun 16 10:30 /etc/shadow
```

Part B — Creating and Verifying Users (20 Minutes)

Create the following users and ensure that a home directory is created for each:

alice, bob, charlie

```
┌──(kali㉿kali)-[~]
└─$ sudo useradd -m bob

┌──(kali㉿kali)-[~]
└─$ sudo useradd -m charlie

┌──(kali㉿kali)-[~]
└─$ sudo useradd -m alice
```

Verify that all three accounts have been added to /etc/passwd.

```
┌──(kali㉿kali)-[~]
└─$ grep -E '^(alice|bob|charlie):' /etc/passwd
bob:x:1002:1002::/home/bob:/bin/sh
charlie:x:1003:1003::/home/charlie:/bin/sh
alice:x:1004:1004::/home/alice:/bin/sh
```

Determine the UID assigned to each newly created user.

```
┌──(kali㉿kali)-[~]
└─$ id  alice
uid=1004(alice) gid=1004(alice) groups=1004(alice)

┌──(kali㉿kali)-[~]
└─$ id bob
uid=1002(bob) gid=1002(bob) groups=1002(bob)

┌──(kali㉿kali)-[~]
└─$ id charlie
uid=1003(charlie) gid=1003(charlie) groups=1003(charlie)
```

Determine the home directory and default shell assigned to alice.

```
┌──(kali㉿kali)-[~]
└─$ getent passwd alice
alice:x:1004:1004::/home/alice:/bin/sh
```

Inspect /home and verify that the required home directories were created.

```
┌──(kali㉿kali)-[~]
└─$ ls -ld /home/alice /home/bob /home/charlie
drwx------ 5 alice   alice   4096 Sep  3 08:28 /home/alice
drwx------ 5 bob     bob     4096 Sep  3 08:28 /home/bob
drwx------ 5 charlie charlie 4096 Sep  3 08:28 /home/charlie
```

Set passwords for all three accounts.

```
┌──(kali㉿kali)-[~]
└─$ sudo passwd alice
New password:
Retype new password:
passwd: password updated successfully

┌──(kali㉿kali)-[~]
└─$ sudo passwd bob
New password:
Retype new password:
passwd: password updated successfully

┌──(kali㉿kali)-[~]
└─$ sudo passwd charlie
New password:
Retype new password:
passwd: password updated successfully
```

Switch from your current account to alice using a login shell. Verify the change using an appropriate command, and then return to your original account.

```
┌──(kali㉿kali)-[~]
└─$ su - alice
Password:
$ whoami
alice
$ pwd
/home/alice
$ id
uid=1004(alice) gid=1004(alice) groups=1004(alice)
$ exit

┌──(kali㉿kali)-[~]
└─$ whoami
kali
```

Part C — Group Management and Access Control (30 Minutes)

Create two groups: security and developers

```
┌──(kali㉿kali)-[~]
└─$ sudo groupadd security
[sudo] password for kali:

┌──(kali㉿kali)-[~]
└─$ sudo groupadd developers

┌──(kali㉿kali)-[~]
└─$ getent group security
security:x:1005:

┌──(kali㉿kali)-[~]
└─$ getent group developers
developers:x:1006:
```

Add alice and charlie to the security group.

```
┌──(kali㉿kali)-[~]
└─$ sudo usermod -aG security alice

┌──(kali㉿kali)-[~]
└─$ sudo usermod -aG security charlie
```

Add bob to the developers group.

```
┌──(kali㉿kali)-[~]
└─$ sudo usermod -aG developers bob
```

Without switching accounts, verify the group membership of all three users.

```
┌──(kali㉿kali)-[~]
└─$ id alice
uid=1004(alice) gid=1004(alice) groups=1004(alice),1005(security)

┌──(kali㉿kali)-[~]
└─$ id bob
uid=1002(bob) gid=1002(bob) groups=1002(bob),1006(developers)

┌──(kali㉿kali)-[~]
└─$ id charlie
uid=1003(charlie) gid=1003(charlie) groups=1003(charlie),1005(security)

┌──(kali㉿kali)-[~]
└─$ groups alice
alice : alice security

┌──(kali㉿kali)-[~]
└─$ groups bob
bob : bob developers

┌──(kali㉿kali)-[~]
└─$ groups charlie
charlie : charlie security
```

Display the entries corresponding to security and developers from /etc/group.

```
┌──(kali㉿kali)-[~]
└─$ grep -E '^(security|developers):' /etc/group
security:x:1005:alice,charlie
developers:x:1006:bob
```

Add bob to the security group without removing him from the developers group.

```
┌──(kali㉿kali)-[~]
└─$ sudo usermod -aG security bob
```

Verify that bob now belongs to both groups.

```
┌──(kali㉿kali)-[~]
└─$ id bob
uid=1002(bob) gid=1002(bob) groups=1002(bob),1005(security),1006(developers)

┌──(kali㉿kali)-[~]
└─$ grep -E '^(security|developers):' /etc/group
security:x:1005:alice,charlie,bob
developers:x:1006:bob
```